

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Design Webpage using HTML5 & CSS3

2. AIM

To design and develop a webpage using HTML5 and CSS3, applying CSS3 styling techniques such as the box model, Flexbox layout, colours, gradients, and hover transitions, in order to understand how an external stylesheet controls the visual presentation of a semantic HTML5 structure.

3. OBJECTIVE

- To understand how an external CSS3 stylesheet is linked to an HTML5 document using the <link> tag.
 - To apply the CSS box model (margin, padding, border-box) to reset and structure page elements.
 - To use Flexbox (display: flex) to arrange the header, navigation bar, and the main/sidebar layout.
 - To style semantic elements, forms, and native input controls using type and attribute selectors.
 - To apply background gradients, colors, box-shadow, and border-radius for visual enhancement.
 - To use the :hover pseudo-class and CSS transitions to create interactive effects on links and buttons.
 - To observe how CSS3 transforms the visual presentation of a semantic HTML5 page compared to the browser's default styling.
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4. THEORY

CSS3 (Cascading Style Sheets, Level 3) is the styling language used to control how an HTML document looks — its layout, colours, fonts, and visual effects. An external CSS file is linked to an HTML5 page using the <link rel="stylesheet" href="..."> tag inside the <head> section, keeping content (HTML) and design (CSS) separate.

Every HTML element is treated as a box made up of content, padding, border, and margin — this is called the CSS box model. Using box-sizing: border-box makes sizing easier by

including padding and border within the element's total width and height, instead of adding extra space on top.

Flexbox (display: flex) is used to arrange elements in a row or column easily. Properties like justify-content and align-items control how elements are aligned, while the flex property decides how much space each element takes — for example, flex: 7 and flex: 3 to create a 70%/30% layout split.

CSS selectors help target specific elements to style them — element selectors (header, nav), ID selectors (#home), class selectors (.container), and attribute selectors (input[type="text"]). The :hover pseudo-class applies a style only when the user moves the mouse over an element, and transition makes that style change happen smoothly instead of instantly.

Overall, CSS3 takes a plain HTML5 structure and transforms it into a properly styled, visually organized webpage — without changing the actual HTML content.

5. ALGORITHM / PROCEDURE

1. Create a new HTML file and add <!DOCTYPE html> at the top.
 2. Set up the <head> with meta charset, title, and a <link> to lab2.css.
 3. Build the <header> with a heading and <nav> links.
 4. Add a hero <section id="home"> with a heading, text, and a link.
 5. Create a <div class="container"> to hold the main layout.
 6. Inside it, add <main> with <article> blocks for content.
 7. Add a <form> with fieldset, legend, and text/email/select/textarea fields plus submit and reset buttons.
 8. Add an <aside> sidebar with unordered and ordered lists.
 9. Add a <footer> with copyright text.
 10. Create the lab2.css file and reset default margin/padding.
 11. Style the header and nav using Flexbox for alignment.
 12. Style the hero section with a gradient background and centered text.
 13. Style the container, main, and aside using flex values to form a two-column layout.
 14. Style form fields and add hover/transition effects on links and buttons.
 15. Save both files and open the HTML file in a browser to check the styling.
 16. Test the page's layout, hover effects, and form behaviour, and note down the results.
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6. PROGRAM

Exp2.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>ANITS - Anil Neerukonda Institute of Technology & Sciences</title>
  <link rel="stylesheet" href="lab2.css">
</head>
<body>

  <!-- Header Section -->
  <header>
    <h1>ANITS</h1>
    <nav>
```

```

    <a href="#home">Home</a>
    <a href="#articles">Departments</a>
    <a href="#about">About</a>
    <a href="#contact">Contact Us</a>
  </nav>
</header>

<!-- Hero / Showcase Section -->
<section id="home">
  <h2>Anil Neerukonda Institute of Technology & Sciences</h2>
  <p>A premier engineering institution in Visakhapatnam, focused on academic excellence,
research, and strong industry-oriented learning.</p>
  <p><a href="#articles"><b>Explore Our Departments Below &darr;</b></a></p>
</section>

<!-- Main Content Layout -->
<div class="container">

  <!-- Left Side: Main Content Column -->
  <main id="articles">

    <article>
      <h2>1. Computer Science Engineering</h2>
      <p>Established with modern labs and industry-aligned curriculum</p>
      <p>The CSE department focuses on core programming, data structures, AI/ML, and
software development, preparing students for top tech careers through strong placement
support.</p>
      <p>Students get hands-on exposure to real-world projects using tools like
<code>Python</code>, <code>Java</code>, and cloud platforms.</p>
    </article>

    <article>
      <h2>2. Student Enquiry Form</h2>
      <p>For admissions and general queries</p>
      <p>Prospective students and parents can use the form below to reach out to the ANITS
admissions office for any information regarding courses, fees, or campus facilities.</p>
      <br>
      <h3>Contact ANITS</h3>
      <form id="contact" action="#" method="post">
        <fieldset>
          <legend><b>Admission Enquiry Form</b></legend>
          <p>
            <label for="name">Full Name:</label>
            <input type="text" id="name" name="username" required placeholder="John
Doe">
          </p>
          <p>
            <label for="email">Email Address:</label>
            <input type="email" id="email" name="useremail" required>
          </p>
        </fieldset>
      </form>
    </article>
  </main>
</div>

```

```

        <p>
            <label for="topic">Enquiry Topic:</label>
            <select id="topic" name="usertopic">
                <option value="admissions">Admissions</option>
                <option value="academics">Academic Curriculum</option>
                <option value="placements">Placements</option>
            </select>
        </p>
        <p>
            <label for="message">Message Details:</label><br>
            <textarea id="message" name="usermessage" rows="5" cols="40"
placeholder="Type your query here..."></textarea>
        </p>
        <p>
            <input type="submit" value="Submit Enquiry">
            <input type="reset" value="Clear Entries">
        </p>
    </fieldset>
</form>
</article>

</main>

<!-- Right Side: Sidebar Column -->
<aside id="about">
    <h3>About ANITS</h3>
    <p>ANITS is known for its strong academic reputation, modern infrastructure, and
excellent placement record across top companies.</p>

    <h4>Key Highlights:</h4>
    <ul>
        <li>Experienced faculty</li>
        <li>Well-equipped labs</li>
        <li>Strong placement cell</li>
    </ul>

    <h4>Departments Offered:</h4>
    <ol>
        <li>Computer Science Engineering</li>
        <li>Electronics & Communication</li>
        <li>Mechanical Engineering</li>
    </ol>
</aside>

</div>

<!-- Footer Section -->
<footer>
    <p>&copy; 2026 ANITS - Anil Neerukonda Institute of Technology & Sciences. All Rights
Reserved.</p>

```

```
</footer>

</body>
</html>
```

Exp2.css

```
/* ===== Reset & Base ===== */
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: 'Segoe UI', Arial, sans-serif;
  background-color: #f9f9f9;
  color: #333;
  line-height: 1.6;
}

/* ===== Header ===== */
header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  background-color: #1e3c72;
  color: white;
  padding: 15px 30px;
}
header h1 { font-size: 24px; }

nav a {
  color: white;
  text-decoration: none;
  margin-left: 20px;
  font-weight: bold;
  transition: 0.3s;
}
nav a:hover { color: #ffcc00; }

/* ===== Hero Section ===== */
#home {
  text-align: center;
  padding: 50px 20px;
  background: linear-gradient(to right, #2a5298, #4facfe);
  color: white;
}
#home h2 { font-size: 28px; margin-bottom: 15px; }
#home p { max-width: 600px; margin: 0 auto 10px auto; }
#home a { color: #ffcc00; text-decoration: none; }
```

```

/* ===== Main Layout ===== */
.container {
    display: flex;
    gap: 25px;
    padding: 30px;
    align-items: flex-start;
}
main {
    flex: 7;
    background-color: white;
    padding: 25px;
    border-radius: 10px;
    box-shadow: 2px 2px 10px rgba(0,0,0,0.1);
}
aside {
    flex: 3;
    background-color: #f4f4f4;
    padding: 20px;
    border-radius: 10px;
}

/* ===== Articles ===== */
article {
    margin-bottom: 25px;
    padding-bottom: 20px;
    border-bottom: 1px solid #ddd;
}
article h2 { color: #1e3c72; margin-bottom: 8px; }
article p { margin-bottom: 10px; }
code {
    background-color: #eee;
    padding: 2px 6px;
    border-radius: 4px;
    font-family: monospace;
}

/* ===== Form ===== */
form fieldset {
    border: 1px solid #ccc;
    border-radius: 8px;
    padding: 20px;
}
form legend { padding: 0 10px; color: #1e3c72; }
form label { display: inline-block; width: 130px; font-weight: bold; }
form input[type="text"],
form input[type="email"],
form select,
form textarea {
    width: 60%;

```

```

padding: 8px;
margin: 6px 0;
border: 1px solid #ccc;
border-radius: 6px;
}
form input[type="submit"],
form input[type="reset"] {
padding: 8px 18px;
margin-top: 10px;
margin-right: 10px;
border: none;
border-radius: 6px;
background-color: #1e3c72;
color: white;
cursor: pointer;
}
form input[type="submit"]:hover,
form input[type="reset"]:hover { background-color: #4facfe; }

/* ===== Sidebar ===== */
aside h3 { color: #1e3c72; margin-bottom: 10px; }
aside h4 { margin-top: 15px; margin-bottom: 8px; }
aside ul, aside ol { margin-left: 20px; }

/* ===== Footer ===== */
footer {
text-align: center;
padding: 18px;
background-color: #1e3c72;
color: white;
margin-top: 20px;
}

```

7. CODE EXPLANATION

Tag / Property	Purpose
<link>	Links external CSS file to HTML
<header>	Top section with title and nav
<nav>	Holds navigation links
<section id="home">	Hero/intro section
<div class="container">	Wraps main + sidebar layout
<main>	Main content area
<article>	Individual content block
<form>	Collects user input
<fieldset>, <legend>	Groups and labels form fields
<input>	Text, email, submit, reset fields
<select>	Dropdown menu
<textarea>	Multi-line text input
<aside>	Sidebar content

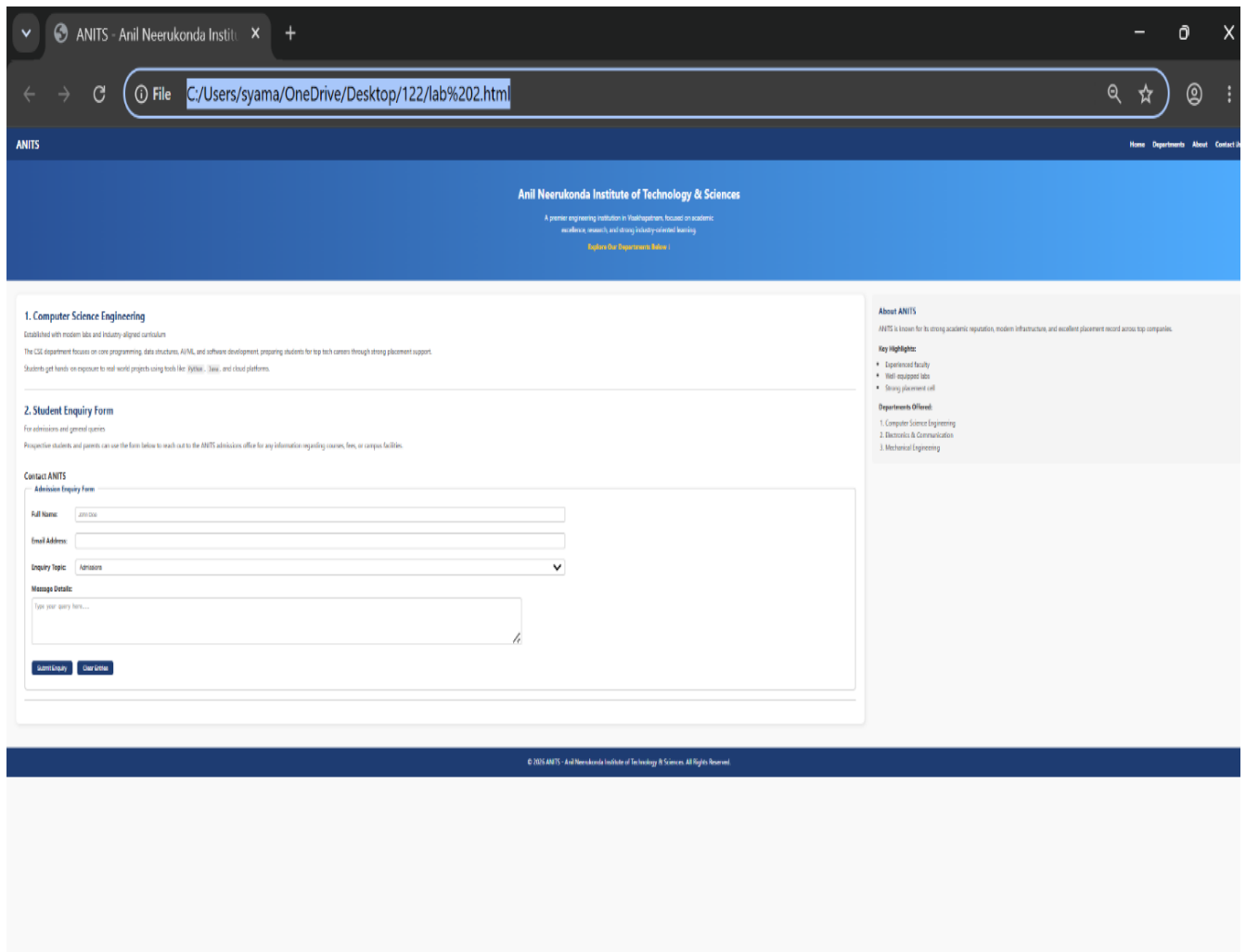
<footer>	Bottom copyright section
display: flex	Arranges items in a row/column
justify-content	Aligns items along main axis
linear-gradient()	Creates gradient background
:hover	Applies style on mouse-over
transition	Animates style changes smoothly
box-sizing: border-box	Includes padding/border in element size
border-radius	Rounds element corners

8. TEST CASES AND EXECUTION DATA

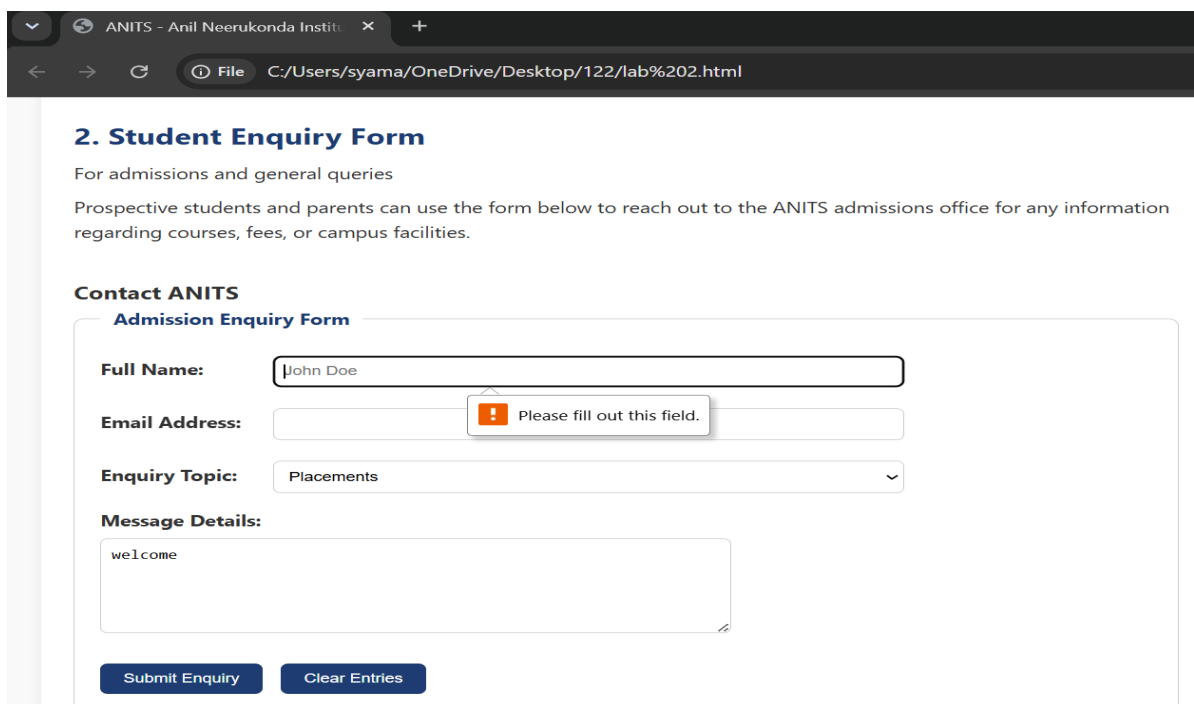
Test Case	Action	Expected Result	Actual Result	Status
TC-01	Open HTML file in browser	Page renders with header, hero gradient, two-column layout, footer	Rendered correctly with all CSS3 styles applied	Pass
TC-02	Hover over a nav link	Link colour turns gold	Colour changed to gold smoothly	Pass
TC-03	Leave "Full Name" empty, click Submit	Shows required-field error	Error shown, form not submitted	Pass
TC-04	Hover over Submit button	Background turns light blue	Colour changed as expected	Pass
TC-05	Fill valid data, click Submit	Form submits with no errors	Submitted successfully	Pass

9. OUTPUT

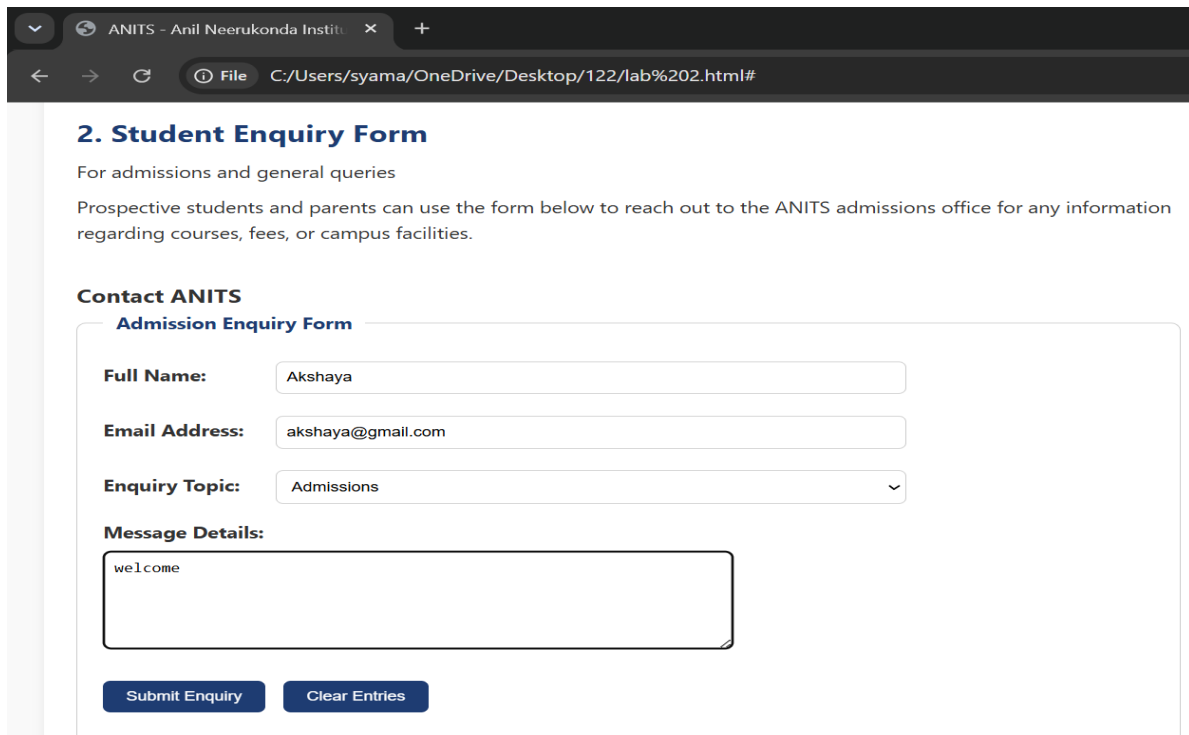
Output 1: Webpage Rendered in Browser



Output 2: Required Field Validation



Output 3: Successful Form Submission

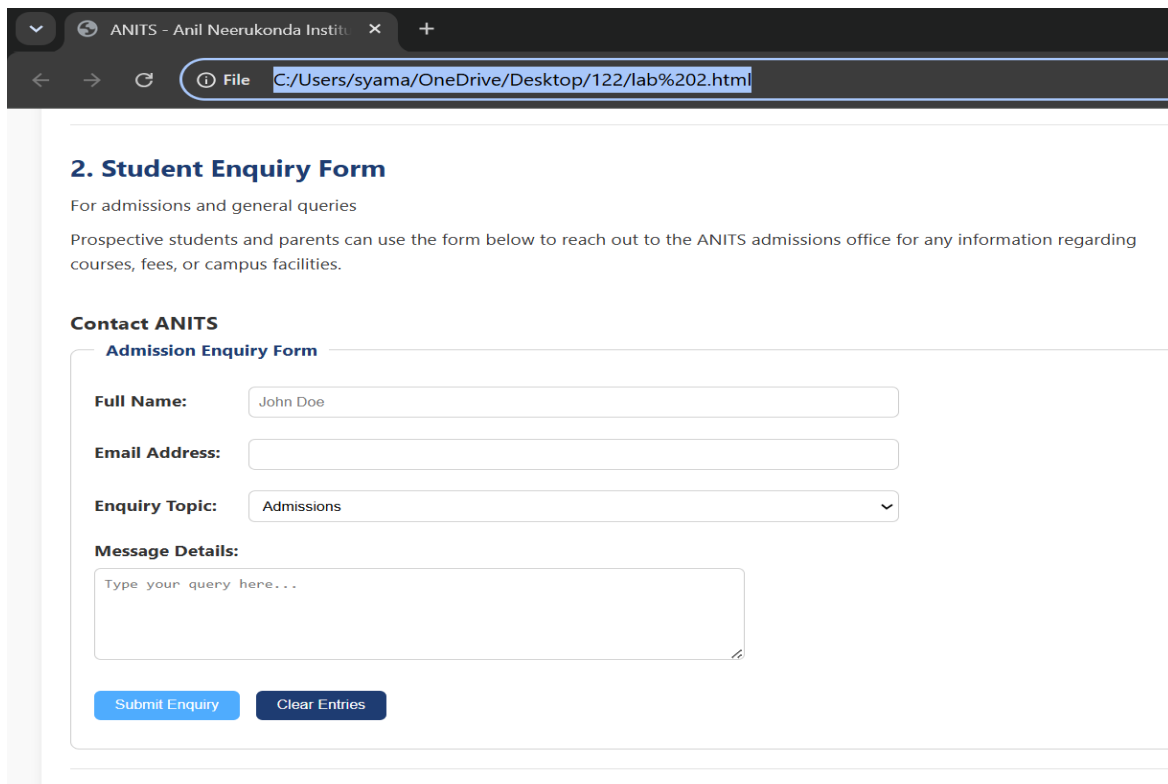


The screenshot shows a web browser window with the address bar displaying the file path: C:/Users/syama/OneDrive/Desktop/122/lab%202.html#. The page title is "ANITS - Anil Neerukonda Institi...". The main heading is "2. Student Enquiry Form". Below the heading, there is a sub-heading "Contact ANITS" and a section titled "Admission Enquiry Form". The form contains the following fields:

- Full Name:** Akshaya
- Email Address:** akshaya@gmail.com
- Enquiry Topic:** Admissions (selected from a dropdown menu)
- Message Details:** welcome

At the bottom of the form, there are two buttons: "Submit Enquiry" and "Clear Entries".

Output 4: Hover over Submit button



The screenshot shows the same web browser window as in Output 3. The address bar displays the file path: C:/Users/syama/OneDrive/Desktop/122/lab%202.html. The page title is "ANITS - Anil Neerukonda Institi...". The main heading is "2. Student Enquiry Form". Below the heading, there is a sub-heading "Contact ANITS" and a section titled "Admission Enquiry Form". The form contains the following fields:

- Full Name:** John Doe
- Email Address:** (empty)
- Enquiry Topic:** Admissions (selected from a dropdown menu)
- Message Details:** Type your query here...

At the bottom of the form, there are two buttons: "Submit Enquiry" and "Clear Entries". The "Submit Enquiry" button is highlighted with a blue border, indicating it is the focus of the hover effect.

10. RESULT

Thus, a webpage was successfully designed and developed using HTML5 and CSS3, applying the box model, Flexbox layout, gradients, and hover transitions to style a semantic HTML structure. The webpage was executed and tested in a web browser, and all test cases — covering page rendering, hover effects, required-field validation, button interaction, and successful form submission — were verified and found to work correctly.